

HPY413Assignment: 7

Network traffic monitoring using the Snort intrusion detection system

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Part 3:

1. Student's Packet Rule (TCP to Port 54321)

Rule: alert tcp 192.168.100.1 any -> 192.168.1.1 54321 (msg:"Detected TCP traffic to port 54321"; sid:1000001;)

Description: This rule detects TCP packets that are sent from IP 192.168.100.1 to destination IP 192.168.1.1 on port 54321. The alert message indicates that a TCP connection was attempted on port 54321.

2. Student's Packet Detected Rule (TCP to Port 54321 with Payload "maria-12345678")

Rule: alert tcp any any -> 192.168.1.1 54321 (msg:"Student's Packet Detected"; content:"maria-12345678"; sid:1000001; rev:1;)

Description: This rule looks for TCP packets sent to IP 192.168.1.1 on port 54321 with the content "maria-12345678" in the payload. The rule is designed to capture custom packets sent to a specific port and payload.

3. Port Scan Packets Rules (Multiple Services)

Rule: alert tcp any any -> 192.168.1.2 80 (msg:"HTTP Port Scan Detected"; content:"maria-12345678"; sid:1000002; rev:1;)

alert tcp any any -> 192.168.1.2 443 (msg:"HTTPS Port Scan Detected"; content:"maria-12345678"; sid:1000003; rev:1;)

alert tcp any any -> 192.168.1.2 22 (msg:"SSH Port Scan Detected"; content:"maria-12345678"; sid:1000004; rev:1;)

alert tcp any any -> 192.168.1.2 23 (msg:"Telnet Port Scan Detected"; content:"maria-12345678"; sid:1000005; rev:1;)

alert tcp any any -> 192.168.1.2 21 (msg:"FTP Port Scan Detected"; content:"maria-12345678"; sid:1000006; rev:1;)

alert udp any any -> 192.168.1.2 53 (msg:"DNS Port Scan Detected"; content:"maria-12345678"; sid:1000007; rev:1;)

alert tcp any any -> 192.168.1.2 554 (msg:"RTSP Port Scan Detected"; content:"maria-12345678"; sid:1000008; rev:1;)
alert tcp any any -> 192.168.1.2 1433 (msg:"SQL Port Scan Detected"; content:"maria-12345678"; sid:1000009; rev:1;)
alert tcp any any -> 192.168.1.2 3389 (msg:"RDP Port Scan Detected"; content:"maria-12345678"; sid:1000010; rev:1;)
alert tcp any any -> 192.168.1.2 1883 (msg:"MQTT Port Scan Detected"; content:"maria-12345678"; sid:1000011; rev:1;)

Description: These rules detect a port scanning attempt on multiple services (HTTP, HTTPS, SSH, Telnet, FTP, DNS, RTSP, SQL, RDP, MQTT) by looking for packets with the content "maria-12345678". Each service is monitored individually, ensuring that if one of these ports is targeted, an alert is generated.

4. Base64 Malicious Packets Rule (TCP to Port 8080)

Rule: alert tcp any any -> 192.168.1.3 8080 (msg:"Base64 Malicious Packet Detected"; content:"MTMyNDU2Nzg5MA=="; sid:1000012; rev:1;)

Description: This rule detects TCP packets sent to port 8080 with a Base64-encoded payload (MTMyNDU2Nzg5MA==). This rule is useful for identifying attempts to send potentially harmful payloads to web services running on port 8080.

5. DNS Suspicious Domain Packet Rule

Rule: alert udp any any -> any 53 (msg:"Suspicious DNS Query Detected"; content:"malicious.example.com"; sid:1000013; rev:1)

Description: This rule detects suspicious DNS queries made to port 53 for the domain "malicious.example.com".

6. Ping Test Packet Rule (ICMP)

Rule: alert icmp any any -> 192.168.1.4 any (msg:"Ping Test Detected"; content:"PingTest-2024"; sid:1000014; rev:1;)

Description: This rule captures ICMP packets (Ping requests) sent to IP 192.168.1.4 with the content "PingTest-2024". The rule ensures these tests are logged for analysis.

7. Snort Rule to Detect Slammer Worm Traffic

Rule: alert udp any any -> any 1434 (msg:"Slammer Worm Detected"; content:"|04 01|"; depth:2; sid:1000020; rev:1;)

Description: This rule is specifically designed to detect traffic associated with the Slammer Worm, a well-known attack targeting port 1434 on Microsoft SQL servers. The worm spreads rapidly by exploiting a vulnerability and uses specific UDP patterns. The rule detects the malicious payload in the traffic (|04 01|), helping to identify the attack in real-time.

By running in terminal the following command: `sudo snort -r /home/student/Desktop/assignment7/snort/lab/ custom_traffic.pcap -c /etc/snort/snort.conf -A console`, the alerts created are displayed below:

```
Commencing packet processing (pid=15506)
12/27-19:56:22.976292  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:80
12/27-19:56:22.976468  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:443
12/27-19:56:22.976664  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:22
12/27-19:56:22.976826  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:23
12/27-19:56:22.976974  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:21
12/27-19:56:22.977119  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:53
12/27-19:56:22.977263  [**] [1:503:7] MISC Source Port 20 to <1024 [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.100.2:20 -> 192.168.1.2:554
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```